

EARTHLAB 2026

Earth Data Analytics Research Network

Programme Schedule — 15–19 June 2026

Department of Engineering, University of Cambridge

Organisers: Francesco Muschitiello (Geography) & Alice Cicirello (Engineering)

Mon–Wed: 9:00–17:00 | Thu–Fri: 10:00–16:00

Lunch provided on Monday, Wednesday, and Friday

Monday 15 June

9:00–17:00 — Introductions, AI Tools, and Data Fundamentals | Oatley Meeting Room, Engineering

Time	Session	Lead
09:00–09:15	Welcome, logistics, and week overview	Francesco & Alice
09:15–10:15	Participant presentations: research, data, and research question	All
10:15–10:30	Coffee break	
10:30–12:00	Introduction to Claude Code and AI agentic workflows for research	Francesco
12:00–12:30	Practical: applying Claude Code to research tasks	Participants
12:30–13:30	Lunch (served)	
13:30–15:00	Lecture: Data types, preparation, and fundamentals of machine learning	Alice
15:00–15:15	Coffee break	
15:15–16:30	Practical on data preparation using own research data	Participants
16:30–17:00	Wrap-up and Q&A	Francesco & Alice

Overnight task: prepare a short presentation on data preparation results for Tuesday morning.

Tuesday 16 June

9:00–17:00 — Neural Networks: ANNs and CNNs | Oatley Meeting Room, Engineering

Time	Session	Lead
09:00–09:30	Short presentations from Monday practical	Participants
09:30–11:00	Lecture: Artificial Neural Networks	Alice
11:00–11:15	Coffee break	
11:15–12:30	Individual project on ANN	Participants
12:30–13:30	Lunch (own)	
13:30–15:00	Lecture: Convolutional Neural Networks	Alice
15:00–15:15	Coffee break	
15:15–16:30	Workgroup on CNN	Participants
16:30–17:00	Wrap-up and preparation for Wednesday presentations	All

Overnight task: finalise CNN workgroup results for presentation on Wednesday morning.

Wednesday 17 June

9:00–17:00 — Bayesian Methods and Project Kick-off | Oatley Meeting Room, Engineering

Time	Session	Lead
09:00–09:30	Presentations from Tuesday workgroup	Participants
09:30–11:00	Lecture: Bayesian statistics and MCMC applied to climate data	Francesco
11:00–11:15	Coffee break	
11:15–12:30	Workgroup on Bayesian statistics	Participants
12:30–13:30	Lunch (served)	
13:30–14:30	Presentations from Bayesian workgroup	Participants
14:30–14:45	Coffee break	
14:45–17:00	Individual project work begins	Participants

Evening: pub outing (optional)

Thursday 18 June

10:00–16:00 — Individual Project Day | Oatley Meeting Room, Engineering

Time	Session	Lead
10:00–12:30	Individual project work + one-on-one sessions	All
12:30–13:30	Lunch (own)	
13:30–16:00	Individual project work + one-on-one sessions	All

Participants work on their own research projects using the methods learned during the week. Francesco and Alice rotate through one-on-one consultations to provide targeted support.

Friday 19 June

10:00–16:00 — Presentations and Closing | Seminar Room, Geography (TBC)

Time	Session	Lead
10:00–11:00	Final project work and presentation preparation	Participants
11:00–11:15	Coffee break	
11:15–13:00	Participant presentations (~15 min each + 5 min Q&A)	All
13:00–14:00	Lunch (served)	
14:00–15:45	Participant presentations (continued)	All
15:45–16:00	Closing remarks and next steps	Francesco & Alice

Notes

Presentations

Monday AM (5 min each): Introduce yourself, your research, your data, and the question you want to tackle this week.

Tuesday and Wednesday AM: Report back on the previous day's practical or workgroup.

Friday (20 min each): Present your individual project work from the week.

Workgroups and Practicals

Each lecture is followed by a hands-on session — either an individual practical or a small-group workgroup — where participants apply the methods just introduced. Workgroups are done in groups of 3–4, with each group receiving a different dataset. Groups present their results the following morning.

Individual Project Work (Wednesday PM – Friday AM)

From Wednesday afternoon onwards, participants work on their own research projects using the methods learned during the week. Francesco and Alice provide one-on-one support on Thursday.

Teaching Datasets

Datasets for the group exercises will be provided via a shared folder. Participants should bring a workable subset of their own research data (under 1 GB) on their laptops.

Software Setup

Participants will receive setup instructions in advance. Required: Miniconda, Git, Claude Code (with API key), and Slack.

Lunch and Refreshments

Lunch is provided on Monday, Wednesday, and Friday. On Tuesday and Thursday, participants should make their own arrangements. Coffee and refreshments will be available throughout.

Pub Outing

An informal group outing is planned for Wednesday evening. Details to follow.